

Rational-Flux Anisotropic Harper Dynamics: Chambers Phase Collapse and the Algebraic Bloch-Band Atlas

HCS-C371 / HEN-O355

4 September 2026

Abstract

For every reduced rational magnetic flux and every positive hopping anisotropy, we fix one magnetic-cell phase convention and derive the exact Chambers characteristic identity. The proof exposes why the vertical transfer trace has only three Laurent modes and fixes both phase signs. We then recover the full two-dimensional spectrum as a polynomial preimage, give the exact algebraic band-contact criterion, prove Aubry duality, flux reversal and parity, and isolate the forced central contact at every even denominator. The one- and two-site cells are computed separately.

1 Magnetic owner and phase convention

Let $(p, q) = 1$, $q \geq 3$, and $\lambda > 0$. On $\ell^2(\mathbb{Z}^2)$ consider the bounded self-adjoint operator

$$(\mathbf{H}_{p/q, \lambda} \Psi)_{m, n} = \Psi_{m+1, n} + \Psi_{m-1, n} + \lambda e^{2\pi i p m / q} \Psi_{m, n+1} + \lambda e^{-2\pi i p m / q} \Psi_{m, n-1}. \quad (1)$$

Fourier transformation in the second coordinate and magnetic Bloch reduction give the q -dimensional fiber

$$(\mathbf{H}(k_x, k_y)u)_m = u_{m+1} + u_{m-1} + 2\lambda \cos\left(k_y + \frac{2\pi p m}{q}\right) u_m, \quad u_{m+q} = e^{iqk_x} u_m. \quad (2)$$

Thus $X = qk_x$ is the *total* horizontal boundary phase. This distinction is essential: throughout,

$$D_{p/q, \lambda}(E; k_x, k_y) := \det(EI_q - \mathbf{H}_{p/q, \lambda}(k_x, k_y)). \quad (3)$$

This manuscript's first revision is the **Chambers phase-collapse owner**.

2 The Chambers polynomial

Put $\zeta = e^{2\pi i / q}$, $y = e^{ik_y}$, and

$$A_m(E, y) = \begin{pmatrix} E - \lambda(y\zeta^{pm} + y^{-1}\zeta^{-pm}) & -1 \\ 1 & 0 \end{pmatrix}, \quad M_q = A_{q-1} \cdots A_0. \quad (4)$$

Theorem 1 (Anisotropic Chambers identity). *There is a unique real monic polynomial $\mathcal{P}_{p/q, \lambda}$ of degree q such that*

$$D_{p/q, \lambda}(E; k_x, k_y) = \mathcal{P}_{p/q, \lambda}(E) - 2\cos(qk_x) - 2\lambda^q \cos(qk_y) \quad (5)$$

for every real E, k_x, k_y .

Proof. Since $\det A_m = 1$, the Floquet multipliers of M_q are $e^{\pm iqk_x}$. Both sides below are monic of degree q and have the same roots, hence

$$D_{p/q, \lambda} = \text{tr } M_q - 2\cos(qk_x). \quad (6)$$

Replacing y by $\zeta^p y$ cyclically permutes the potential sequence in (4). The monodromy changes by cyclic conjugacy, so its trace is invariant. Because p is invertible modulo q , a Laurent polynomial of y with degrees between $-q$ and q and this invariance can use only the modes $-q, 0, q$.

The coefficient of y^q has one contribution: select $-\lambda y \zeta^{pm}$ in every upper-left entry. It equals

$$(-\lambda)^q \zeta^{pq(q-1)/2} = -\lambda^q. \quad (7)$$

Indeed, odd q makes $q-1$ even, while even q forces p odd. Complex conjugation gives the same coefficient for y^{-q} . The remaining mode is independent of both phases, real, and monic; call it $\mathcal{P}_{p/q,\lambda}(E)$. Equations (6) and (7) prove (5). Monicity gives uniqueness. $\square$

The theorem is convention-complete: using the boundary character X in place of k_x rewrites the horizontal term as $-2 \cos X$, not $-2 \cos(qX)$. This is precisely the convention tested by the executable fibers.

3 Spectrum and algebraic band contacts

This revision is the **spectrum, duality, and edge owner**. Write

$$\mathcal{C}_{q,\lambda} = 2(1 + \lambda^q), \quad B_{p/q,\lambda}(E) = \mathcal{P}_{p/q,\lambda}(E)^2 - \mathcal{C}_{q,\lambda}^2. \quad (8)$$

Lemma 1 (Sourced cyclic-continuant evaluation). *If q is even, then*

$$\mathcal{P}_{p/q,\lambda}(0) = 2(-1)^{q/2}(1 + \lambda^q). \quad (9)$$

Proof. Lamoureux and Mingo use $h_{\theta,L} = u + u^{-1} + (L/2)(v + v^{-1})$. Their Theorem 2.5 proves the vanishing of every intermediate cyclic-matching sum, and their Corollary 2.6 gives, for even q ,

$$\Delta_{p/q,L}(0) = 2(-1)^{q/2}(1 + (L/2)^q) \quad [5].$$

Our diagonal potential is $2\lambda \cos(\cdot)$, so their parameter is $L = 2\lambda$. Comparing the phase-independent monic term in their Chambers formula with Theorem 1 identifies $\Delta_{p/q,2\lambda} = \mathcal{P}_{p/q,\lambda}$: their defining vertical phase $\pi/(2q)$ kills the explicit q th cosine, while our transfer trace then equals $\mathcal{P}$. This also fixes our $\det(EI - H)$ convention. Substitution gives (9). Thus this all- q step is sourced rather than inferred from the finite cyclotomic audit. $\square$

Theorem 2 (Complete Bloch spectral atlas). *The full two-dimensional spectrum of (1) is*

$$\text{Spec}(H_{p/q,\lambda}) = \{E \in \mathbb{R} : |\mathcal{P}_{p/q,\lambda}(E)| \leq \mathcal{C}_{q,\lambda}\}. \quad (10)$$

The two edge factors are actual fiber characteristic polynomials:

$$\mathcal{P}(E) - \mathcal{C}_{q,\lambda} = D(E; 0, 0), \quad \mathcal{P}(E) + \mathcal{C}_{q,\lambda} = D(E; \pi/q, \pi/q). \quad (11)$$

Consequently the algebraic band-edge multiset is the real zero multiset of B . An edge label is multiple precisely when

$$\mathcal{P}(E) = \pm \mathcal{C}_{q,\lambda}, \quad \mathcal{P}'(E) = 0. \quad (12)$$

Moreover,

$$\mathcal{P}_{p/q,\lambda}(E) = \lambda^q \mathcal{P}_{p/q,1/\lambda}(E/\lambda), \quad (13)$$

$$\mathcal{P}_{p/q,\lambda}(E) = \mathcal{P}_{(q-p)/q,\lambda}(E), \quad (14)$$

$$\mathcal{P}_{p/q,\lambda}(-E) = (-1)^q \mathcal{P}_{p/q,\lambda}(E). \quad (15)$$

If q is even, then

$$\mathcal{P}_{p/q,\lambda}(0) = 2(-1)^{q/2}(1 + \lambda^q), \quad \mathcal{P}'_{p/q,\lambda}(0) = 0, \quad (16)$$

so zero is a multiple central edge.

Proof. Magnetic Bloch theory is a direct-integral decomposition into (2). By (5), E lies in some fiber spectrum exactly when

$$\mathcal{P}(E) = 2 \cos(qk_x) + 2\lambda^q \cos(qk_y).$$

The phases vary independently, and their sum fills $[-\mathcal{C}_{q,\lambda}, \mathcal{C}_{q,\lambda}]$, proving (10). Putting $(k_x, k_y) = (0, 0)$ and $(\pi/q, \pi/q)$ in (5) proves (11). Both endpoint fibers are real symmetric: their total horizontal boundary phases are 0 and π , and their diagonal potentials are real. Thus both edge factors have only real roots, with multiplicity, and $B = (\mathcal{P} - \mathcal{C})(\mathcal{P} + \mathcal{C})$

is real-rooted. Equation (12) follows from $B' = 2\mathcal{P}\mathcal{P}'$ and $\mathcal{P} \neq 0$ at an edge. It is an algebraic coalescence criterion, not a claim that every remaining gap is open.

Exchange of the two lattice axes reverses the magnetic orientation and swaps the hopping amplitudes. After division by λ , magnetic Fourier duality identifies the result with $\lambda H_{(q-p)/q, 1/\lambda}$. Complex conjugation identifies the fluxes p/q and $(q-p)/q$. Comparing the phase-independent monic terms in (5) proves (13)–(14).

The bipartite involution $\Gamma\Psi_{m,n} = (-1)^{m+n}\Psi_{m,n}$ conjugates H to $-H$ and shifts both quasimomenta by π . Substitution in (5) proves (15). For even q , Lemma 1 gives the first formula in (16). Equation (15) makes $\mathcal{P}$ even, so $\mathcal{P}'(0) = 0$. This proves the derivative formula without extrapolating the finite exact lane. $\square$

Proposition 1 (Accumulated small-cell boundaries). *When coincident wrap neighbors are added rather than overwritten, Theorem 1 has the direct boundary polynomials*

$$q = 1, p = 0 : \quad \mathcal{P}(E) = E, \quad q = 2, p = 1 : \quad \mathcal{P}(E) = E^2 - 2(1 + \lambda^2). \quad (17)$$

The two $q = 2$ folded bands meet at zero. As $\lambda \downarrow 0$, the spectrum tends to the decoupled horizontal-chain interval $[-2, 2]$; reciprocal duality is not evaluated at $\lambda = 0$.

Proof. For $q = 1$ the scalar fiber is $2\cos k_x + 2\lambda\cos k_y$. For $q = 2$, write $X = e^{2ik_x}$ and $Y = e^{ik_y}$. The off-diagonal entries are $1 + X^{-1}$ and $1 + X$, while the diagonal entries are $\pm\lambda(Y + Y^{-1})$. Its determinant is

$$E^2 - 2(1 + \lambda^2) - (X + X^{-1}) - \lambda^2(Y^2 + Y^{-2}),$$

which proves (17) and the asserted contact. $\square$

References

- [1] P. G. Harper, *Single band motion of conduction electrons in a uniform magnetic field*, Proc. Phys. Soc. A 68 (1955), 874–878.
- [2] W. G. Chambers, *Linear-network model for magnetic breakdown in two dimensions*, Phys. Rev. 140 (1965), A135–A143.
- [3] D. R. Hofstadter, *Energy levels and wave functions of Bloch electrons in rational and irrational magnetic fields*, Phys. Rev. B 14 (1976), 2239–2249.
- [4] J. Avron, P. H. M. van Mouche, and B. Simon, *On the measure of the spectrum for the almost Mathieu operator*, Comm. Math. Phys. 132 (1990), 103–118.
- [5] M. P. Lamoureux and J. A. Mingo, *On the characteristic polynomial of the almost Mathieu operator*, Proc. Amer. Math. Soc. 135 (2007), 3205–3215. <https://doi.org/10.1090/S0002-9939-07-08830-2>.
- [6] S. Jitomirskaya, L. Konstantinov, and I. Krasovsky, *On the spectrum of critical almost Mathieu operators in the rational case*, arXiv:2007.01005.